Supplementary material

Forecasting Dissolved Nitrous Oxide in Wastewater Treatment
with a Hybrid Deep Learning Probabilistic Model

Table S1: Trainable-parameter accounting for the structurally matched multi-scale comparison.

Component	Matched BiLSTM	SWBiLSTM
One 64/32-unit branch	75,264	37,760
Three branches	225,792	113,280
Dense fusion head	14,465	14,465
Total	240,257	127,745
Reduction vs. matched BiLSTM	–	46.83%

The parameter reduction is calculated relative to the structurally matched multi-scale BiLSTM:

$$\frac{240{,}257 - 127{,}745}{240{,}257} \times 100\% \approx 46.83\%.$$

GPR training cases and kernel hyperparameters are excluded from the trainable neural parameter counts.

Table S2: Audited data flow and chronological partitions used in the repeated experiments.

Item	Count or value	Description
Raw records	831,557	All rows in the source file.
Finite timestamp–target records	508,801	Records remaining after invalid timestamps and non-finite targets were removed.
Removed invalid/non-finite records	322,756	No artificial imputation was applied.
Modal sampling interval	120 s	Nominal interval used in the continuity audit.
Nonmodal or nonpositive transitions	6,818	Discontinuities identified in the finite chronological series.
Longest audited span	101 steps	Corresponding to the 72-step history and the 30-step maximum horizon.
Strictly continuous common targets	448,065	Target timestamps available under the same continuity rule at every horizon.
Eligible fraction	88.06%	Fraction of finite records retained as common targets.
Training targets	313,645	First chronological 70% of eligible targets.
Validation targets	22,403	Used for early stopping and residual-gain selection.
Calibration targets	22,403	Reserved for interval-calibration sensitivity analysis.
Testing targets	89,614	Final chronological holdout used only for evaluation.
First test target	2024-01-23 18:18 UTC	Identical first target time at all four horizons.
Scaler-fit observations	363,860	Observations preceding the first validation target.
Scaler mean	0.112008	Training-only target mean in mg-N L^{-1}.
Scaler standard deviation	0.270027	Training-only population standard deviation in mg-N L^{-1}.

The scaler-fit count refers to finite observations preceding the first validation target, whereas the partition counts refer to continuity-eligible forecast targets. The two counts therefore describe different sets of observations.

Table S3: Uncalibrated 95% prediction-interval performance of SWBiLSTM–GPR on the common test set, before the optional split-conformal adjustment. Values are mean ± sample standard deviation across five seeds. MPIW is expressed in mg-N L^{-1}.

Horizon	PICP	PINAW	MPIW
2 min	0.9644 ± 0.0164	0.0348 ± 0.0068	0.0721 ± 0.0140
10 min	0.9723 ± 0.0092	0.0949 ± 0.0100	0.1964 ± 0.0207
30 min	0.9633 ± 0.0134	0.1708 ± 0.0203	0.3536 ± 0.0421
60 min	0.9606 ± 0.0123	0.2354 ± 0.0228	0.4873 ± 0.0471

Table S4: GPR residual-correction settings and optimized kernel quantities. Values are mean ± sample standard deviation across five seeds. The subset size was fixed at 500 training cases for every run.

Horizon	Subset size	Residual gain a	Kernel constant	Length-scale	Noise level
2 min	500	0.728 ± 0.125	2.341 ± 1.613	2.192 ± 1.770	0.00362 ± 0.00534
10 min	500	0.664 ± 0.107	1.506 ± 0.185	1.159 ± 0.387	0.00667 ± 0.00783
30 min	500	0.220 ± 0.094	1.484 ± 0.075	1.443 ± 0.439	0.00538 ± 0.00331
60 min	500	0.351 ± 0.047	1.637 ± 0.141	1.271 ± 0.548	0.02344 ± 0.02347

GPR implementation details.

For each forecast horizon and seed, the GPR model was fitted to in-sample residuals from the trained SWBiLSTM backbone using 500 stratified training cases. The 71 input features were standardized using statistics estimated from this subset only.

For the selected training cases, the kernel matrix was defined by

$$K_{ij} = C\, k_{5/2}(\mathbf{z}_i, \mathbf{z}_j; \ell) + \sigma_n^2 \delta_{ij},$$

where $k_{5/2}$ denotes the isotropic Matérn kernel with smoothness parameter $5/2$, C is the kernel constant, ℓ is a scalar length-scale shared across the standardized input dimensions, and σ_n^2 is the white-noise variance. The Kronecker delta δ_{ij} equals one when $i = j$ and zero otherwise.

The initial values were $C = 1$, $\ell = 1$, and $\sigma_n^2 = 10^{-2}$. Optimization bounds were $[10^{-3}, 10^{3}]$ for both C and ℓ, and $[10^{-6}, 10^{1}]$ for σ_n^2. Kernel parameters were fitted by maximizing the log-marginal likelihood using the default L-BFGS-B optimizer. The setting `n_restarts_optimizer=0` specified one optimization run from the initial parameters without additional restarts. An additional fixed diagonal term of 10^{-6} was included during fitting through `alpha=1e-6`.

The setting `normalize_y=True` internally centered and scaled the residual response before fitting. Consequently, the kernel constant and noise variance in Table S4 refer to the internally normalized residual scale. The reported noise level is a variance, whereas the quantity following ± is its sample standard deviation across the five runs.

The residual gain a was selected separately by minimizing validation squared error over $[0, 1]$. It scaled the residual-mean correction only and did not rescale the predictive standard deviation. All kernel parameters were fitted using the selected training cases; the validation-selected gain was then fixed before test evaluation.

Table S5: Computational cost at the 2-min forecast horizon. Runtime values are mean ± sample standard deviation across five seeds. GPR fitting used 500 stratified training cases. Inference values are batch-averaged recorded prediction times per test sample; the hybrid entry combines neural and GPR prediction times and excludes feature preparation and deployment overheads.

Model or stage	**Trainable neural parameters**	**Neural training (s)**	**GPR fitting (s)**	**Inference (ms/sample)**
LSTM	33,665	88.6 ± 27.2	–	0.0111 ± 0.0024
Matched multi-scale BiLSTM	240,257	162.0 ± 20.6	–	0.0317 ± 0.0011
SWBiLSTM	127,745	160.3 ± 29.3	–	0.0285 ± 0.0008
SWBiLSTM-GPR	127,745	160.3 ± 29.3	1.03 ± 0.32	0.0800 ± 0.0063

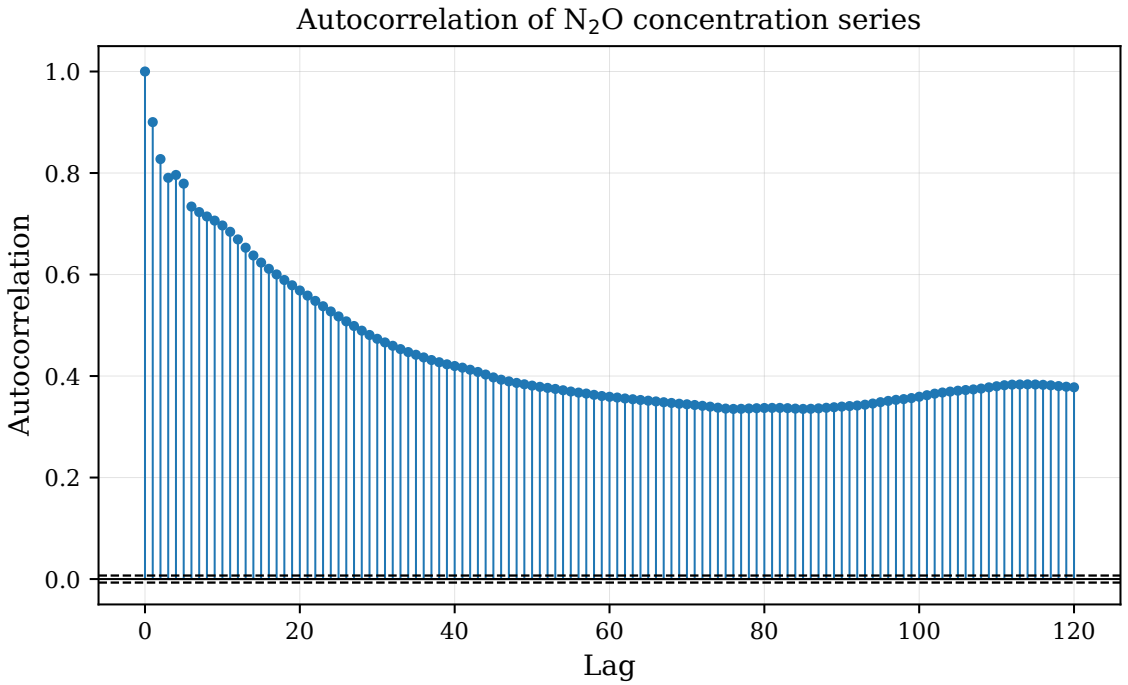


Figure S1: Autocorrelation analysis of the dissolved N_2O concentration series.